THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 07

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Student A is correct	$3 \times 5^2 = 75$; B took the product
2	**(c)** Incorrect — the factorisation is unique (FTA)	Fundamental Theorem of Arithmetic
3	**(b)** $3, -7$	$(x-3)(x+7)$
4	**(b)** 3	$\frac{3}{k} = 1$
5	**(a)** infinitely many solutions	All ratios equal $\frac{5}{3}$
6	**(b)** Incorrect — it has 0, 1 or infinitely many solutions	Two distinct lines meet at most once
7	**(d)** $(x+2)^2 = x^2 + 3$	Reduces to $4x + 1 = 0$
8	**(b)** $x^2 + 6x - 40 = 0$	$(x+3)^2 = 49$
9	**(b)** 99	$4 + 19 \times 5$
10	**(d)** 400	$n^2 = 400$
11	**(d)** $\frac{2}{3}$	Ratio of corresponding sides
12	**(c)** Incorrect — their angles need not be equal	e.g. $70^\circ, 70^\circ, 40^\circ$ vs $50^\circ, 50^\circ, 80^\circ$
13	**(a)** $\frac{1}{2}$	$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$
14	**(a)** 4 cm	$\sqrt{25 - 9}$
15	**(b)** 44 cm	$\frac{1}{3} \times 132$
16	**(b)** 36 cm	$\pi r + 2r = 22 + 14$
17	**(a)** 18π cm³	$\frac{2}{3}\pi \times 27$
18	**(a)** 13	$45 - 32$
19	**(b)** 7	Middle value
20	**(b)** $\frac{3}{13}$	$\frac{12}{52}$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	No common prime factor ⇒ HCF $= 1$; LCM $= 2^3 \times 3^2 \times 5^2 = 1800$	1+1
22	$\frac{1}{2} + \frac{3}{2} + k = 0 \Rightarrow k = -2$; other zero $= -\frac{3}{2} - \frac{1}{2} = -2$	1+1
23	$16 - 4p = 0 \Rightarrow p = 4$	1+1
24	$\frac{2.4}{DB} = \frac{3.2}{4.8} \Rightarrow DB = 3.6$ cm	1+1
25	$\left(\frac{7+2}{3}, \frac{9+6}{3}\right) = (3, 5)$	1+1
26	Divide by $\cos\theta$: $\frac{5\tan\theta - 3}{5\tan\theta + 2} = \frac{4-3}{4+2} = \frac{1}{6}$	1+1
27	$\frac{50\sqrt{3}}{\sqrt{3}} = 50$ m	1+1
28	$PT = 15$ cm; $\angle OTP = 90^\circ$	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,4), (2,0)$; $(0,1), (-1,0)$. Intersect at $(1,2)$ ⇒ consistent	2+1
30	$n = 40$; cf 8, 20 ⇒ class 10–20: $10 + \frac{20-8}{12} \times 10 = 20$	1+1+1
31	$105, \ldots, 994$: $\frac{994-105}{7} + 1 = 128$	1+1+1
32	Each such chord is bisected at the point of contact: length $= 2\sqrt{R^2 - r^2}$, the same for all	1+1+1
33	Area 235.5 cm²; arc 31.4 cm; perimeter 61.4 cm	1+1+1
34	$\frac{13}{25}, \frac{8}{25}, \frac{9}{25}$	1+1+1
35	Distances $75\sqrt{3}$ and 75 m ⇒ $75(\sqrt{3} - 1) \approx 54.9$ m	1+1+1

	OR: $h = x\sqrt{3} = \frac{80-x}{\sqrt{3}} \Rightarrow x = 20$; $h = 20\sqrt{3} \approx 34.64$ m; distances 20 m and 60 m	1+1+1
36	$\Sigma f = 50$, $\Sigma fx = 100 + 320 + 720 + 1600 + 300 + 240 = 3280$ ⇒ mean $= 65.6$. The student is wrong.	1+1+1
	OR: Classes 140–145: 7, 145–150: 18, 150–155: 11, … $\frac{n}{2} = 25.5$ ⇒ class 145–150: $145 + \frac{25.5-11}{18} \times 5 = 149.03$ cm	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Right angle at $(8, -2)$ (one side vertical, one horizontal) ⇒ circumcentre = mid-point of hypotenuse $= (5, 2)$ (3); radius $= 5$ (2)	3+2
	OR: (i) $\frac{3k-2}{k+1} = 0 \Rightarrow k = \frac{2}{3}$ ⇒ 2 : 3; point $(0, 1)$ (ii) $\sqrt{25 + 100} = 5\sqrt{5}$	3+2
38	$\frac{1}{\sec A - \tan A} = \sec A + \tan A$ and $\frac{1}{\sec A + \tan A} = \sec A - \tan A$ ⇒ both sides equal $\tan A$	2+2+1
	OR: LHS $= \csc^2 A \sec^2 A = \frac{1}{\sin^2 A \cos^2 A} = \frac{1}{\sin^2 A(1 - \sin^2 A)}$	2+2+1
39	(i) $500 - 89.83 = 410.17$ cm³ (ii) $400 - 38.5 + 77 = 438.5$ cm²	3+2
	OR: (i) $314 + 261.67 = 575.67$ cm³ (ii) $l = 13$: $204.1 + 157 = 361.1$ cm²	3+2
40	Median: cf 6, 16 ⇒ $200 + \frac{25-16}{15} \times 50 = 230$. Mode: $200 + \frac{15-10}{30-10-12} \times 50 = 231.25$	3+2
	OR: (i) $a = 75$: $\Sigma fu = 14$ ⇒ 77.33 h (ii) cf 6, 16, 34 ⇒ $70 + \frac{14}{18} \times 10 = 77.78$ h (iii) $70 + \frac{8}{36-26} \times 10 = 78$ h	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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